An investigation into the relationship between air pollution and asthma and Type 1 diabetes based on UK data

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MSc Data Science

Project Proposal

An investigation into the relationship between air pollution and asthma and Type 1 diabetes based on UK data

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Abstract

The proposal is for a project to investigate health data relating to two diseases and data relating to air pollution. The diseases are Asthma and Type 1 Diabetes. The health data is collected from the UK National Health Service digital portal. The pollution data is air pollution data collected by the UK Department of Environment Food and Rural Affairs and from the London Air Quality Network. The project will use geographical data to enhance the analysis. Visualisations of the data will be produced. The data will be investigated for correlations and clusters.

Table of Contents

1. **Introduction** **4**
2. **Objectives** **4**
3. **Background** **4**
4. **Data Sources** 5
5. **Approach** 5
   - 5.a Data Preparation 6
   - 5.b Visualisation 6
6. **Software** **7**
7. **Analysis** **7**
8. **Plan** **8**
9. **References** **8**

1. Introduction

This proposal is for a Data Science project to explore and visualise data on two diseases (which have been associated with environmental pollution) and air pollution. The project will gather data on air pollution in the UK and combine it with epidemiological data on both diseases. Data Science tools will then be used to analyse the data and methods developed to visualise the data and the results of the analysis. This project will require both a software design and implementation effort and a statistical modelling and analysis exercise.

The diseases that will be considered are Asthma and Type 1 Diabetes. Asthma is a disease of the lungs and airways. It is thought that its causes are both environmental and genetic. A significant environmental factor is air quality. Air quality is thought to play a part in the incidence of Asthma by causing inflammation to the lungs and airways. Air quality is affected by several types of pollution including smoking, industrial pollution and traffic pollution. There is medical literature describing the mechanisms of how airborne pollution can cause Asthma. Type 1 Diabetes is an auto-immune disease that affects the production of insulin. It is thought that it is caused by a combination of genetic and environmental factors. The causal connection between Type 1 Diabetes and air pollution is not well established. The UK NHS accurately monitors Asthma and Type 1 Diabetes cases. Data about cases of both diseases at NHS surgeries (practices) is made available by NHS Digital as part of the Qualities and Outcomes Framework (QOF).

The air pollution that will be considered is measured by road traffic air quality monitors. The pollution data will come from statistics published by the Department for Environment, Food and Rural Affairs (DEFRA) and from the London Air Quality Network (Ref 1) project run by King's College London. DEFRA manages networks of monitoring stations which monitor airborne pollution levels. Their monitoring covers the whole of the UK. The London Air Quality Network manages monitoring sites for London.

To the best of my knowledge no attempt has so far been made to analyse the co-occurrence in geography of air pollution levels and these two disease occurrences for the whole of England and Wales.

2. Objectives

The objectives of the project are to:

1. Develop geographical and statistical visualisations of the data
2. Discover if there are significant correlations between traffic pollution and the auto-immune diseases (asthma and Type 1 diabetes) based on the datasets
3. Discover if asthma and Type 1 diabetes are correlated
4. Use clustering algorithms on the data to identify possible geographic clusters of the diseases and pollution hotspots

3. Background

Two recent studies that consider the connection between Asthma and pollution are:

1. Ambient Air Pollution and Adult Asthma Incidence in Six European Cohorts (ESCAPE) Environ Health Perspect. 2015 Jun (Ref 2)
2. Effect of Early Life Exposure to Air Pollution on Development of Childhood Asthma Environ Health Perspect. 2010 Feb (Ref 3)

Paper 1 used pollution data from 714 sites in Europe in 2001. It used a more complex Geographical Information System for determining the air pollution values than I will be using.

Paper 2 was based on a study of all children born in southwestern British Columbia in 1999 and 2000. Instead of birth data I shall be using admissions to GP Surgeries as the health data component of my research.

In the course of the project further research will be undertaken to gather more data on the epidemiology of the diseases and to enhance the available data on air pollution.

4. Data Sources

Medical data and the pollution data for corresponding time frames will be used. However, there will need to be aggregation of the data in order to produce datasets that correspond. Most of the data is in spreadsheet form. The medical data is held at the level of GP practices in a matrix of approximately 7,000 rows representing practices (surgeries). There are more than 50 columns of data for each practice from which the relevant data needs to be extracted. Occurrences of diseases are given as percentages of the population.

The practice details include geographical codes which are defined by the government statistics agency. Corresponding with these codes are geographical boundaries defined by co-ordinates. These are defined in https://data.gov.uk/dataset/national-statistics-postcode-lookup-uk.

The pollution data can be downloaded from the DEFRA site as csv files. There are 248 monitoring sites in England and Wales. However not every site is always providing data. Also there are geographical gaps in the coverage. There are almost 100 different pollutants that are recorded. The data is available for different time frames. Averages for the pollution data will have to be computed so that the data will correspond to the health data.

5. Approach

I will develop software to:

1. Read the health files, filter and calculate the relevant data
2. Read the national statistics postcode lookup file for the co-ordinates of the geographic boundaries
3. Create a dataset of health data and boundaries
4. Read the pollution data files
5. Apply algorithms to model pollution

6. Aggregate the pollution data so that it corresponds with the timeframes of the health data
7. Create a combined dataset of health and pollution data
8. Create visual maps (including choropleth maps) of the health regions with displays of the disease occurrences for each region and the corresponding pollution levels.
9. Create statistical models of the dataset to understand significant predictors for the diseases
10. Analyse the data to determine if there are clusters of the disease and produce heat maps

5.1 Data Preparation

The aim of the data preparation will be to combine relevant fields from the input files in order to create a dataset that is suitable for visualisation and statistical analysis. The health data will be reduced to the fields required for the analysis. Location codes in the health data will then be combined with geographical data. The locations of the pollution monitoring stations will be used to calculate the corresponding GP practices. Selected pollution data will then be added to the dataset.

An algorithm will be applied to the pollution data to model the contribution of areas of high pollution to neighbouring areas. I will consider using the Gravity Model to compute the effect that distance has on the distribution of pollution. A version of the Gravity model was developed to predict trade flows between two trading partners based on the size of their economies and the distance between them. It has also been used to model the effects of trans national air pollution, see Air Pollution in Europe 2020: The Gravity Model and EKC Decomposition, Ulla Lehmijoki and Elena Rovenskaya (Ref 4).

5.2 Visualisation

The visualisation will use Choropleth maps. These are maps which use different colours within geographic boundaries to show the proportion of a variable. An example is shown in Fig 1 below. This shows England and Wales divided into regions that correspond to Clinical Commissioning Groups (CCG). CCGs are administrative zones that contain GP surgeries. An approach will be developed to enable both the pollution data and health data to be visualised on the map. This will probably require the map to be layered so that both factors can be visualised.

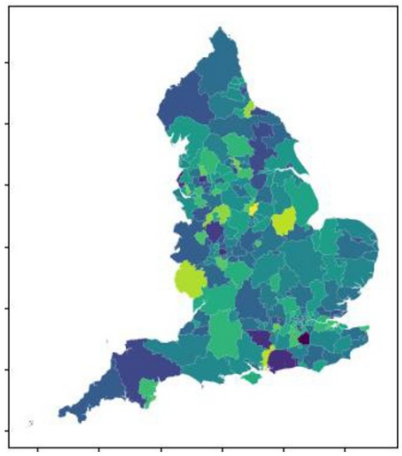

Figure 1. A choropleth map of Clinical Commissioning Groups

Heatmaps will also be developed to display the results of the data analysis. An example is shown in Figure 2. This is a display of air pollution in London taken from the London Air Quality Network.

Figure 2. A heatmap of pollution in London

6. Software and Methodology

I am planning to develop the software using Python. The following Python libraries will be useful:

Pandas – facilitates the reading of files and provides DataFrames for data manipulation

GeoPandas – makes working with geospatial data easier

Scikit-learn- for regression and clustering analysis

7. Analysis

1. Correlation
   In order to discover any correlation between pollution and disease occurrence I will run Poisson Regressions on the dataset. This form of regression analysis is used to model count data. I will also run a Logistic Regression. For this the occurrences of disease will be converted into a classifier and the dataset adjusted appropriately.
2. Clusters
   I will also examine the data for cluster of disease and pollution using cluster analysis algorithms
3. Confounding factors
   I will explore the impact of confounding factors e.g. smoking. Asthma UK has some interesting visualisations at https://www.asthma.org.uk/get-involved/campaigns/data-visualisations/ ,which show that prevalence is most common in cities in the North of England. However I had assumed that traffic pollution would be worse in London. In this case there may be other factors that lead to the increased disease incidence.

8. Plan

1. Further research on available data from April to May
2. Develop software to prepare the data from May to June
3. Run analysis from June to July
4. Create visualisations of results from August to September

9. References

1. London Air Quality Network https://www.londonair.org.uk/LondonAir/Default.aspx
2. Ambient Air Pollution and Adult Asthma Incidence in Six European Cohorts (ESCAPE) Environ Health Perspect. 2015 Jun
3. Effect of Early Life Exposure to Air Pollution on Development of Childhood Asthma Environ Health Perspect. 2010 Feb
4. Air Pollution in Europe 2020: The Gravity Model and EKC Decomposition, Ulla Lehmijoki and Elena Rovenskaya